



2 Real Analysis 2023

Tutorial 2

A/Prof Elena Berdysheva

Dr Francesco Russo



Suprema, infima, maxima, minima

Problem 4 Let $A \subseteq \mathbb{R}$ be a nonempty set.

- (a) Assume that $\max A$ exists. Show that $\max A = \sup A$.
- (b) Assume that A is bounded above, and $\sup A \in A$. Show that $\max A$ exists and $\max A = \sup A$.
- (c) We denote $-A = \{-a : a \in A\}$. Assume that A is bounded above. Show that $-A$ is bounded below and

$$\inf(-A) = -\sup A.$$

- (d) Assume that A is bounded below with $\inf A > 0$. We denote $A^{-1} = \{a^{-1} : a \in A\}$. Show that A^{-1} is bounded above and

$$\sup(A^{-1}) = (\inf A)^{-1}.$$

Solution.

- (a) Let $M = \max A$. By the definition of a maximum, $M \in A$ and $a \leq M$ for all $a \in A$. The latter property means that M is an upper bound for A . Let N be another upper bound for A . Then $M \leq N$, because $M \in A$. It follows that M is the least upper bound for A , i.e. $M = \sup A$.
- (b) Let $M = \sup A$. By the definition of a supremum, $a \leq M$ for all $a \in A$. By assumption, $M \in A$. This is exactly the definition of the maximum, i.e. $M = \max A$.
- (c) Assume that A is bounded above. Let $M = \sup A$. We have that $a \leq M$ for all $a \in A$. Now take $b \in -A$. Then $b = -a$ with some $a \in A$. The inequality $a \leq M$ implies $b = -a \geq -M$. Thus, $-A$ is bounded below and $-M$ is a lower bound for $-A$.

It remains to show that $-M$ is the greatest lower bound of $-A$. Let n be another lower bound for $-A$. This means that $n \leq b$ for all $b \in -A$. Take $a \in A$, then $a = -b$ with some $b \in -A$. We see that $a = -b \leq -n$. It follows that $-n$ is an upper bound for A . Since M is the least upper bound of A , we have $M \leq -n$. This implies $n \leq -M$. The latter inequality shows that $-M$ is the greatest lower bound for $-A$, i.e. $\inf(-A) = -M = -\sup A$.

- (d) Let $m = \inf A > 0$. Then $m \leq a$ for all $a \in A$. It follows, in particular, that all elements of A are positive.

Now let $b \in A^{-1}$, then $b = a^{-1}$ with some $a \in A$. It follows that $b = a^{-1} \leq m^{-1}$. Thus, A^{-1} is bounded above and m^{-1} is an upper bound for A^{-1} .

It remains to show that m^{-1} is the least upper bound of A^{-1} . Let N be another upper bound for A^{-1} . Since all elements of A are positive, also all elements of A^{-1} are positive, and therefore $N > 0$. By the definition of an upper bound, $b \leq N$ for all $b \in A^{-1}$. Let $a \in A$, then $a = b^{-1}$ with some $b \in A^{-1}$. We have $a = b^{-1} \geq N^{-1}$. It follows that N^{-1} is a lower bound for A . Since m is the greatest lower bound, $m \geq N^{-1}$. This implies $m^{-1} \leq N$, i.e. $\sup(A^{-1}) = m^{-1} = (\inf A)^{-1}$.

Problem 5 Determine, without proof, suprema, infima, maxima and minima, if they exist, of the following sets:

- (a) $A = \left\{ \frac{3n}{2n+1} : n \in \mathbb{N} \right\}$,
- (b) $B = \{[1 + (-1)^n]n : n \in \mathbb{N}\}$,
- (c) $C = \left\{ (-1)^n + \frac{1}{2^n} : n \in \mathbb{N} \right\}$,
- (d) $D = \left\{ \frac{2n}{n+m} : n, m \in \mathbb{N} \right\}$.

Solution.

In this problem we expect you to determine suprema, infima, maxima and minima of the given sets without proof. However, below we give not just the answers, but also some explanations.

(a) $A = \left\{ \frac{3n}{2n+1} : n \in \mathbb{N} \right\} = \left\{ 1, \frac{6}{5}, \frac{9}{7}, \frac{12}{9}, \frac{15}{11}, \dots \right\}$.

We see that the elements of A build an increasing sequence that tends to $\frac{3}{2}$. We conclude that $\inf A = \min A = 1$, $\sup A = \frac{3}{2}$, $\max A$ does not exist.

(b) $B = \{[1 + (-1)^n]n : n \in \mathbb{N}\} = \{0, 2, 0, 4, 0, 6, 0, 8, \dots\}$.

The elements in odd positions are all 0. The elements in the even position are even integers that grow to infinity. The set B is not bounded above, and thus $\sup B$ and $\max B$ do not exist. On the other hand, $\inf B = \min B = 0$.

(c) $C = \left\{ (-1)^n + \frac{1}{2^n} : n \in \mathbb{N} \right\} = \left\{ -1 + \frac{1}{2}, 1 + \frac{1}{4}, -1 + \frac{1}{8}, 1 + \frac{1}{16}, -1 + \frac{1}{32}, 1 + \frac{1}{64}, \dots \right\}$.

The elements in the odd positions build a decreasing sequence that tends to -1 . The elements in the even positions build a decreasing sequence that tends to 1. We conclude that $\max C = \sup C = 1 + \frac{1}{4} = \frac{5}{4}$, $\inf C = -1$, $\min C$ does not exist.

(d) $D = \left\{ \frac{2n}{n+m} : n, m \in \mathbb{N} \right\}$.

Since $m > 0$, we have $\frac{2n}{n+m} < \frac{2n}{n} = 2$. Thus, D is bounded above and 2 is an upper bound for D . On the other hand, elements $\frac{2n}{n+1}$ build a sequence that tends to 2

as $n \rightarrow \infty$, and this means that these elements come arbitrarily close to 2. This implies that $\sup D = 2$. On the other hand, $2 = \sup D \notin D$. Indeed, $\frac{2n}{n+m} = 2$ is equivalent to $n = n + m$, and this is not possible for $n, m \in \mathbb{N}$. We conclude that $\max D$ does not exist.

Since $n, m > 0$, we see that $\frac{2n}{n+m} > 0$. Thus D is bounded below and 0 is a lower bound for D . The elements $\frac{2}{1+m}$ build a sequence that tends to 0 as $m \rightarrow \infty$, so that these elements come arbitrarily close to 0. This implies that $\inf D = 0$. Since $0 \notin D$, we conclude that $\min D$ does not exist.

The Archimedean property and the nested interval property

Problem 6 (a) Prove that $\bigcap_{n=1}^{\infty} [n, \infty) = \emptyset$.

(b) Prove that $\bigcap_{n=1}^{\infty} \left(0, \frac{1}{n}\right) = \emptyset$.

(c) Determine $\bigcap_{n=1}^{\infty} \left[0, \frac{1}{n}\right]$.

Solution.

(a) Let $x \in \mathbb{R}$. By the Archimedean property, there exists $n_0 \in \mathbb{N}$ with $n_0 > x$, so that $x \notin [n_0, \infty)$. Since $\bigcap_{n=1}^{\infty} [n, \infty) \subseteq [n_0, \infty)$, we have $x \notin \bigcap_{n=1}^{\infty} [n, \infty)$.

This is true for any $x \in \mathbb{R}$. Consequently, $\bigcap_{n=1}^{\infty} [n, \infty) = \emptyset$.

(b) Let $x > 0$. By the Archimedean property, there exists $n_0 \in \mathbb{N}$ with $0 < \frac{1}{n_0} < x$. Then $x \notin \left(0, \frac{1}{n_0}\right)$. Since $\bigcap_{n=1}^{\infty} \left(0, \frac{1}{n}\right) \subseteq \left(0, \frac{1}{n_0}\right)$, we have $x \notin \bigcap_{n=1}^{\infty} \left(0, \frac{1}{n}\right)$.

The above is true for any $x > 0$. But if $x \leq 0$, then clearly x does not lie in any of the sets $\left(0, \frac{1}{n}\right)$, and therefore also not in their intersection. Consequently, $\bigcap_{n=1}^{\infty} \left(0, \frac{1}{n}\right) = \emptyset$.

(c) We claim that $\bigcap_{n=1}^{\infty} \left[0, \frac{1}{n}\right] = \{0\}$.

Clearly, $0 \in \left[0, \frac{1}{n}\right]$ for every $n \in \mathbb{N}$, and therefore $0 \in \bigcap_{n=1}^{\infty} \left[0, \frac{1}{n}\right]$. To show that the intersection does not contain any further points, we argue as above. Clearly, if $x < 0$, then $x \notin \bigcap_{n=1}^{\infty} \left[0, \frac{1}{n}\right]$. If $x > 0$, then by the Archimedean property there exists

$n_0 \in \mathbb{N}$ with $0 < \frac{1}{n_0} < x$, so that $x \notin \left[0, \frac{1}{n_0}\right]$. It follows that $x \notin \bigcap_{n=1}^{\infty} \left[0, \frac{1}{n}\right]$.